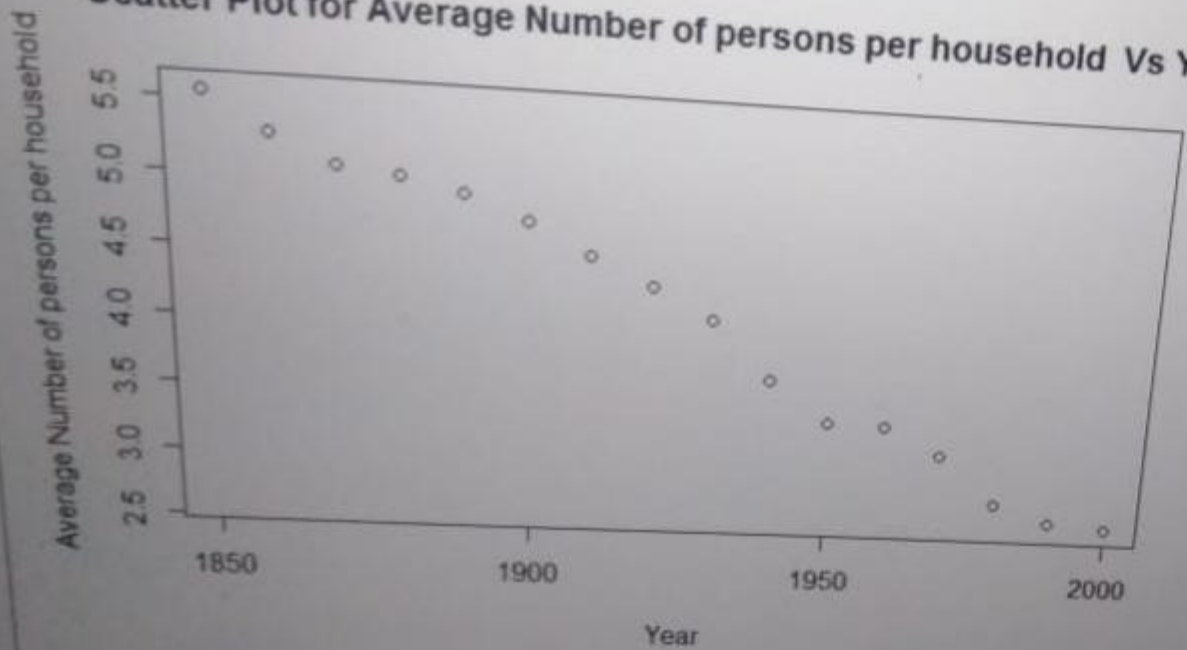


U.S. Census Bureau estimates of the average number of persons per household in the United States for ce
2000 are extracted from the file perhouse on the companion website. Figure below displays the scatter plot

Scatter Plot for Average Number of persons per household Vs Year



R outputs of the regression model are shown below.

Regression Model

Coefficients

Response: Acid content of material.

	Df	Sum Sq	Mean Sq	F value
Organic acid content	A	C	2517.02	G
Residuals	B	262.98	E	
Total	9	D	F	

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Find values marked A, B, C, D, E, F and G in the ANOVA table.

1. A:

2. B:

3. C: (Keep all the decimal places in the answer)

4. D: (Keep all the decimal places in the answer)

5. E: (Keep all the decimal places in the answer)

6. F:

7. G: (Round off the answer up to the fourth decimal point)

(Type your answers within the given spaces except for "F")

i. What can be concluded using the scatter plot?

There is a positive linear relationship in between organic acid content and acid

ii. State the estimated regression equation.

Estimated Organic acid content = $31.7087 + 0.3533$ (Acid content of material)

iii. State in how much acid content of material will change if organic acid content

Acid content of material will increase by 31.7087 units.

iv. Does slope of the Regression line is significant ($\beta \neq 0$)? Use $\alpha = 5\%$.

Yes. R

v. Use the regression equation to predict the acid content of material if organic

82.2306 units

7.00

Flag question

companion website.)

Year	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960	1970	1980	1990	2000
Average number of persons per Household	5.55	5.28	5.09	5.04	4.93	4.76	4.54	4.34	4.11	3.67	3.37	3.35	3.14	2.76	2.63	2.51

Analysis of Variance Table

Response: Average number of persons per household

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Year	A	14.9164	F	G	5.875e-14***
Residuals	B	D	0.0174		
Total	C	E			

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Find values marked A, B, C, D, E, F and G in the ANOVA table.

1. A: 1

2. B: 14

3. C: 15

4. D: 0.2436 (keep all the decimal places in the answer)

5. E: 15.16 (keep all the decimal places in the answer)

6. F: 14.9164 (keep all the decimal places in the answer)

7. G: 857.2644 (round off the answer up to the fourth decimal point)

(Type your answers within the given spaces)

A researcher is interested in finding whether there is any relationship in between temperature and proportion of impurities through solid helium. Temperature is measured in degrees centigrade ($^{\circ}\text{C}$). He used a sample size of 10. The following information is given.

$$\sum x = -2685.6 \quad \sum y = 7.007 \quad \sum xy = -1893.5644 \quad \sum (x^2) = 721454.96 \quad \sum (y^2) = 5.634$$

Where X is the temperature and Y is the proportion of impurities.

Calculate Pearson's correlation coefficient between the two variables (Give your answer in four decimal places).
Type your answer within the given space)

Answer:

Flag question

claim, they selected 40 random patients who were suffering from Back Pain and given this drug. Average re
hours with 2.25 standard deviation. Test the hypothesis at 5% level of significance.

a) Hypothesis : $H_0: \mu \leq 6.5$ Vs $H_1: \mu > 6.5$ ▼

b) Test Statistic (Under H_0) : $Z = (\bar{X} - \mu) / (\sigma / \sqrt{n})$ ▼

c) Distribution of test statistic : Normal Distribution (N (0,1)) ▼

d) Critical value : 1.64 ▼

e) Rejection criteria : Choose... ▼

f) Test value : Choose...

g) Conclusion (In Scientific term) : Reject H_0 if $Z_{cal} > 1.64$ at 5% level of significance

Reject H_0 if $T_{cal} > 1.64$ at 5% level of significance

Reject H_0 if $T_{cal} < 1.64$ at 5% level of significance

Reject H_0 if $Z_{cal} < -1.64$ at 5% level of significance

Reject H_0 if $Z_{cal} > 1.96$ at 5% level of significance

h) Conclusion (In general term) : Choose...

Online Exams

Sri Lanka Institute of Information Technology

A researcher has found that average life time of a truck tire is 50500 miles with a standard deviation of 5000 miles. If a truck tire follows a normal distribution, find the following probabilities,

1. Probability that life time of a truck tire is at least 60000 miles : (Keep all the decimal places)
2. Probability that life time of a truck tire is 45000 to 58900 miles : (Keep all the decimal places)
3. What is the life time (in miles) where 38% of truck tires took less than this life time? mile (integer)

(Type your answers within the given spaces)



Online Exams

Sri Lanka Institute of Information Technology

Question 4

Not yet answered

Marked out of 12.00

🚩 Flag question

A researcher found that pulse rates of women, who are affected by COVID-19 virus has a mean of 75 and Assuming that pulse rate of women follows a normal distribution, find the following probabilities.

1. Probability that a COVID-19 infected woman has less than 70 pulse rate : 0.266 (Keep all the d
2. Probability that a COVID-19 infected woman has pulse rate in between 55 and 90 : 0.9634 (Ke
answer)
3. At which pulse rate, 30% of COVID-19 infected women have more than that pulse rate? 1 (Ke
points)

(Type your answers within the given spaces)



Online Exams

Sri Lanka Institute of Information Technology

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g question

A Continuous random variable X has a probability density function given by,

$$f_X(x) = \begin{cases} k(3x^2 - 3) & ; -1 \leq x \leq 1 \\ 0 & ; \text{otherwise} \end{cases}$$

a) Find k value (Round off your answer to three decimal values) :

b) Find Expected Value (Round off your answer to three decimal values) :

c) Find Variance (Round off your answer to three decimal values) :

d) Find $F_X(0.7)$ (Round off your answer to three decimal values) :

(Type the correct answers within the given spaces)



Online Exams

Sri Lanka Institute of Information Technology

A researcher is interested in finding whether there is any relationship between temperature and the proportion of impurities in solid helium. Temperature is measured in degrees centigrade ($^{\circ}\text{C}$). He used a sample size of 10. The following information is given.

$$\sum x = -2685.6 \quad \sum y = 7.007 \quad \sum xy = -1893.5644 \quad \sum (x^2) = 721454.96 \quad \sum (y^2) = 5.63454$$

Where X is the temperature and Y is the proportion of impurities.

Calculate Pearson's correlation coefficient between the two variables (Give your answer in four decimal places).
(Type your answer within the given space)

Answer:

Question 2

Not yet answered

Marked out of 16.00

Flag question

Fertilizer manufacturing company required to test whether the potassium content in a one fertilizer in the packet cover. In the cover it mentioned as 15.5 g. They took a random sample of 25 fertilizer packets of the sample were calculated as 15.0 g and 2.2. Test the manufacturing company's claim at 5% level of significance.

- a) Hypothesis : $H_0: \mu \geq 15.5$ Vs $H_1: \mu < 15.5$ ▼
- b) Test Statistic : $Z = (\bar{X} - \mu) / (S / \sqrt{n})$ ▼
- c) Distribution of test statistic : Normal distribution $(N(0,1))$ ▼
- d) Critical value : -1.711 ▼
- e) Rejection criteria : Reject H_0 if $T_{cal} < -1.711$ at 5% level of significance. ▼
- f) Test value : -1.14 ▼
- g) Conclusion (In Scientific term) : Since $T_{cal} = -1.14$ is greater than -1.711, we do not reject H_0 at 5% level of significance.
- h) Conclusion (In general term) :
We don't have enough evidence to suggest that actual mean potassium content is less than the mentioned value.

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stion

A company, which produced candy products, required to test whether the average sugar content which is marked on the label. In the label it is marked as 2.5 g. From the previous studies it has a standard deviation of 0.15 and it is normally distributed. A group of researchers conducted a sample of 28 toffees. Sample average sugar content was 2.4 g. Test the hypothesis at 5% level.

a) Hypothesis : Choose...

b) Test Statistic : Choose...

c) Distribution of test statistic : Choose...

d) Critical value : Choose...

e) Rejection criteria : Choose...

f) Test value : Choose...

g) Conclusion (In Scientific term): Choose...

h) Conclusion (In general term) : Choose...

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Online Exams

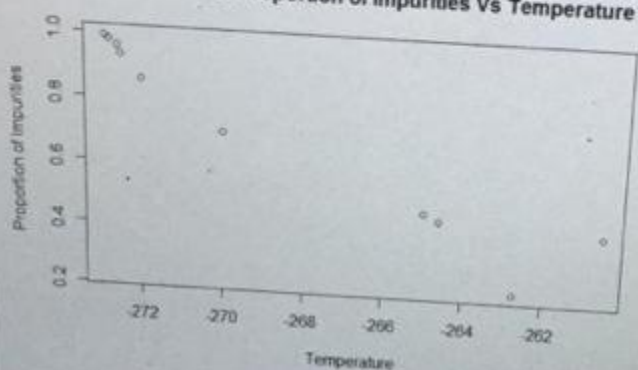
Sri Lanka Institute of Information Technology

8

answered
out of
question

A researcher is interested in finding whether there is any relationship in between temperature and the proportion of impurities passing through solid helium. Temperature is measured in degrees centigrade ($^{\circ}\text{C}$). He used a sample size of 10 to study this relationship. Figure below displays the scatter plot for the data.

Scatter Plot for Proportion of Impurities Vs Temperature



Outputs of the regression model are shown below:



A Continuous random variable X has a probability density function

$$f_X(x) = \begin{cases} k(3x^2 - 3) & ; -1 \leq x \leq 1 \\ 0 & ; \text{otherwise} \end{cases}$$

- a) Find k value (Round off your answer to three decimal values)
- b) Find Expected Value (Round off your answer to three decimal values)
- c) Find Variance (Round off your answer to three decimal values)
- d) Find $F_X(0.7)$ (Round off your answer to three decimal values) I

(Type the correct answers within the given spaces)

Exams

Institute of Information Technology

that average life time of a truck tire is 50500 miles with a standard deviation of 2500 miles.

of a truck tire is at least 60000 miles: (Keep all the decimal places in the answer)

truck tire is 45000 to 58900 miles: (Keep all the decimal places in the answer)

38% of truck tires took less than this life time? miles (Give the answer to the nearest mile)



Online Exams

Sri Lanka Institute of Information Technology

1

answered
out of

question

A typist of a certain Book Publishing Company can type 7 pages per day. Without using any approximation, calculate

a) The typist type more than 8 pages per day

b) The typist type fewer than 3 pages per day

c) Using a suitable approximation, find the probability that the typist type less than or equal to 4 pages per day.

Type your answers within the given spaces [Keep all decimal places of the final answer].

Question 5

Not yet answered

Marked out of
7.00

Flag question

U.S. Census Bureau estimates of the average number of persons per household in the United States in 2000 are shown in the following table with the ANOVA output for the fitted regression model. (The companion website.)

Year	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960	1970	1980
Average number of persons per Household	5.55	5.28	5.09	5.04	4.93	4.76	4.54	4.34	4.11	3.67	3.37	3.35	3.14	2.76

Analysis of Variance Table

Response: Average number of persons per household

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Year	A	14.9164	F	G	5.875e-14***
Residuals	B	D	0.0174		
Total	C	E			

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Find values marked A, B, C, D, E, F and G in the ANOVA table.

1. A:
2. B:
3. C:
4. D: (Keep all the decimal places in the answer)
5. E: (Keep all the decimal places in the answer)
6. F: (Keep all the decimal places in the answer)
7. G: (Round off the answer up to the fourth decimal point)

(Type your answers within the given spaces)

From past records, suppose that on a typical day, 30% of students drive to campus, 60% of the student remaining 10% come to campus in some other way (Eg:- By walk, take the bus, get a ride). The campus encourage people not to drive to campus. After the program, they want to know whether these proportions hypothesis, a random sample of 300 students on a particular day was asked how they got to campus. For

Transportation mode	Drive	Bike	Other	Total
Frequency	100	150	50	300

Do the above results suggest that the past record for proportions have changed? Use 5% level of significance

1. Hypothesis:

Choose...

2. Distribution of test statistic:

Choose...

3. Significance level:

▼

4. Critical value:

Choose... ▼

5. Rejection criteria:

Choose... ▼

6. Test value:

Choose... ▼

7. Conclusion (in Scientific term):

Choose...

8. Conclusion (in General term):

Choose...



Online Exams

Sri Lanka Institute of Information Technology

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ag question

A researcher found that pulse rates of women, who are affected by COVID-19 virus has a mean of 75 and standard deviation is 8. Assuming that pulse rate of women follows a normal distribution, find the following probabilities.

1. Probability that a COVID-19 infected woman has less than 70 pulse rate : (Keep all decimal places in the answer)
2. Probability that a COVID-19 infected woman has pulse rate in between 55 and 90 : (Keep all decimal places in the answer)
3. At which pulse rate, 30% of COVID-19 infected women have more than that pulse rate? (with two decimal points)

(Type your answers within the given spaces)

Total

11

D

F

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i. What can be concluded using the scatter plot?

There is a positive linear relationship in between weekly sales and weekly advertising expenditure. ▼

ii. State the estimated regression equation.

Estimated Weekly sales = 316.025 + 4.241 (Weekly adve

iii. State in how much weekly sales will change if weekly advertising expenditure increased by one un

Weekly sales will increase by 4.241\$. ▼

iv. Does slope of the Regression line is significant ($\beta \neq 0$)? Use $\alpha = 5\%$.

Yes. Regression line is significa

v. Use the regression equation to predict the weekly sales if weekly advertising expenditure is 34\$.



Question 9

Not yet answered

Marked out of 10.00

Flag question

A random group of 40 patients, who were suffered from high blood pressure, has average deviation of 4.5. Construct a 90% confidence interval for the true mean of systolic blood

a) Lower Limit Equation : Choose...

b) Upper Limit Equation : Choose...

c) Critical value : $\bar{X} + Z(10\%) * (S/\sqrt{n})$

$\bar{X} + t(5\%,n) * (S/\sqrt{n})$

d) Lower Limit value : $\bar{X} + Z(5\%) * (S/\sqrt{n})$

$\bar{X} + t(0.10,39) * (S/\sqrt{n})$

$\bar{X} + Z(0.90) * (\sigma/\sqrt{n})$

e) Upper Limit value : Choose...

a) Hypothesis : $H_0: \mu \leq 6.5$ Vs $H_1: \mu > 6.5$ ▼

b) Test Statistic (Under H_0) : $Z = (\bar{X} - \mu) / (\sigma / \sqrt{n})$ ▼

c) Distribution of test statistic : Normal Distribution (N (0,1)) ▼

d) Critical value : 1.64 ▼

e) Rejection criteria : Choose... ▼

f) Test value : Choose... ▼

g) Conclusion (In Scientific term) : Choose...

Choose...

h) Conclusion (In general term) : Since $Z_{cal} = -2.81$ is less than -1.64 reject H_0 at 5% level

Since $Z_{cal} = 2.89$ is greater than 1.96 reject H_0 at 5% level

Since $Z_{cal} = 2.81$ is greater than 1.96 reject H_0 at 5% level

Since $T_{cal} = -2.89$ is less than -1.64 reject H_0 at 5% level

Since $Z_{cal} = 2.81$ is greater than 1.64 reject H_0 at 5% level

16.00

Flag question

claim, they selected 40 random patients who were suffering from Back Pain hours with 2.25 standard deviation. Test the hypothesis at 5% level of significance.

a) Hypothesis : $H_0: \mu \leq 6.5$ Vs $H_1: \mu > 6.5$

b) Test Statistic (Under H_0) : $Z = (\bar{X} - \mu) / (\sigma / \sqrt{n})$

c) Distribution of test statistic : Normal Distribution (N (0,1))

d) Critical value : 1.64

e) Rejection criteria : Choose...

f) Test value : Choose...

g) Conclusion (In Scientific term) : 2.18

-2.81

-2.89

h) Conclusion (In general term) : 2.81

2.89

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estion

Past experience indicates that the time required for high school seniors to complete a standardized test (exam) is a normal distribution with an average of 35 minutes and standard deviation of 4.7 minutes. If a random sample of 20 high school seniors took 32 minutes to complete this exam, test the hypothesis that average completion time (μ) has decreased, at the 0.05 level of significance.

a) Hypothesis : $H_0: \mu \geq 35$ Vs $H_1: \mu < 35$ ▼

b) Test Statistic (Under H_0) : Choose... ▼

c) Distribution of test statistic : Choose... ▼

d) Critical value : 1.96 ▼

e) Rejection criteria : Choose... ▼

f) Test value : Choose... ▼

g) Conclusion (In Scientific term) : Choose... ▼

h) Conclusion (In general term) :
Choose...

Question 10

Not yet answered

Marked out of 1.00

Flag question

A group of researchers wants to investigate the mean relief time given by a certain drug for Back Pain. Past studies have stated that the average relief time is 6.5 hours. But researchers stated that the relief time should be greater than 6.5 hours. To prove their claim, they selected 40 random patients who were suffering from Back Pain and given this drug. Average relief time was 7.25 hours with 2.25 standard deviation. Test the hypothesis at 5% level of significance.

a) Hypothesis : Choose...

b) Test Statistic (Under H_0) : Choose...

c) Distribution of test statistic : Choose...

d) Critical value : Choose...

e) Rejection criteria : Choose...

f) Test value : Choose...

g) Conclusion (In Scientific term) : Choose...

h) Conclusion (In general term) : Choose...

Online Exams
Sri Lanka Institute of Information Technology

U.S. Census Bureau estimates of the average number of persons per household in the United States for census years between 1850 and 2000 are shown in the following table with the ANOVA output for the fitted regression model. (These data are in the file perhouse on the companion website.)

Year	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960	1970	1980	1990	2000
Average number of persons per household	5.55	5.28	5.09	5.04	4.93	4.76	4.54	4.34	4.11	3.67	3.37	3.35	3.14	2.76	2.63	2.59

Analysis of Variance Table

Response: Average number of persons per household

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Year	A	14.9164	F	G	5.875e-14***
Residuals	B	D	0.0174		
Total	C	E			

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Find values marked A, B, C, D, E, F and G in the ANOVA table.

1. A
2. B
3. C
4. D (keep all the decimal places in the answer)
5. E (keep all the decimal places in the answer)
6. F (keep all the decimal places in the answer)
7. G (Round off the answer up to the fourth decimal point)

(Type your answers within the given spaces)



Question 10

Not yet answered

Marked out of
16.00

Flag question

A group of researchers wants to investigate the mean relief time given by a certain drug for Back Pain. A doctor has stated that the average relief time is 6.5 hours. But researchers stated that the relief time should be more than 6.5 hours. To claim, they selected 40 random patients who were suffering from Back Pain and given this drug for 6.5 hours with 2.25 standard deviation. Test the hypothesis at 5% level of significance.

a) Hypothesis : $H_0: \mu \leq 6.5$ Vs $H_1: \mu > 6.5$

b) Test Statistic (Under H_0) : $Z = (\bar{X} - \mu) / (\sigma / \sqrt{n})$

c) Distribution of test statistic : Normal Distribution (N (0, 1))

d) Critical value : 1.64

e) Rejection criteria : Choose...

f) Test value

g) Conclusion (In Scientific term)

h) Conclusion (In general term) : Choose...

Choose...

Reject H_0 if $Z_{cal} > 1.64$ at 5% level of significance

Reject H_0 if $T_{cal} > 1.64$ at 5% level of significance

Reject H_0 if $T_{cal} < -1.64$ at 5% level of significance

Reject H_0 if $Z_{cal} < -1.64$ at 5% level of significance

Reject H_0 if $Z_{cal} > 1.96$ at 5% level of significance

Question 3

Not yet answered

Marked out of
10.00

Flag question

A random group of 40 patients, who were suffered from high blood pressure, had a standard deviation of 4.5. Construct a 90% confidence interval for the true mean of systolic blood pressure.

a) Lower Limit Equation : $\bar{X} - Z(5\%) * (S/\sqrt{n})$

b) Upper Limit Equation : $\bar{X} + Z(5\%) * (S/\sqrt{n})$

c) Critical value : 1.64

d) Lower Limit value : 121.43

e) Upper Limit value : 123.77

The Symbol

$<$

$\leq$

$>$

$\geq$

What It Means

less than
fewer than

less than or
no more than
does not exceed
at most

greater than
more than
exceeds

greater than
at least
no less than



Question 9

Not yet answered

Marked out of 10.00

Flag question

In a classroom of 36 students, has average Statistics marks as 72 with a standard deviation of 5.5. true mean of Statistics marks (μ).

a) Lower Limit Equation : Choose...

b) Upper Limit Equation : Choose...

c) Critical value : Choose...

d) Lower Limit value : Choose...

e) Upper Limit value : Choose...

**Question 10**

Not yet answered

Marked out of 16.00

Flag question

A group of researchers wants to investigate the mean relief time given by a certain drug for Back Pain. It is stated that the average relief time is 6.5 hours. But researchers stated that the relief time should be greater than 6.5 hours. To claim, they selected 40 random patients who were suffering from Back Pain and given this drug. Average relief time was 7.5 hours with 2.25 standard deviation. Test the hypothesis at 5% level of significance.

a) Hypothesis

: Choose...

b) Test Statistic (Under H_0)

:

 $H_0: \mu \geq 6.5$ Vs $H_1: \mu < 6.5$ $H_0: \mu = 6.5$ Vs $H_1: \mu \neq 6.5$ $H_0: \mu = 7.5$ Vs $H_1: \mu \neq 7.5$ $H_0: \mu \leq 7.5$ Vs $H_1: \mu > 7.5$ $H_0: \mu \leq 6.5$ Vs $H_1: \mu > 6.5$

c) Distribution of test statistic

:

d) Critical value

:

Choose...

e) Rejection criteria

:

Choose...

f) Test value

:

Choose...

g) Conclusion (In Scientific term): Choose...

h) Conclusion (In general term): Choose...

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lag question

A group of researchers wants to investigate the mean relief time given by a certain drug for Back Pain. Past studies stated that the average relief time is 6.5 hours. But researchers stated that the relief time should be greater than 6.5 hours. To claim, they selected 40 random patients who were suffering from Back Pain and given this drug. Average relief time was 7.25 hours with 2.25 standard deviation. Test the hypothesis at 5% level of significance.

a) Hypothesis : $H_0: \mu \leq 6.5$ Vs $H_1: \mu > 6.5$

b) Test Statistic (Under H_0) : $Z = (\bar{X} - \mu) / (\sigma / \sqrt{n})$

c) Distribution of test statistic : Normal Distribution (N (0 , 5.06))

d) Critical value : Choose...

e) Rejection criteria : Choose...

f) Test value : Choose...

g) Conclusion (in Scientific term) : Choose...

h) Conclusion (in general term) : Choose...

Question 1

Not yet answered

Marked out of
7.00

Flag question

A study of the amount of rainfall (X) and the quantity of air pollution (Y) shows a positive relationship between the daily rainfall and particulate removed. The regression model is as follows.

Analysis of Variance Table

Response: Particulate Removed

	Df	Sum Sq	Mean Sq	F value	Pr
Rainfall	A	770.26	F	G	4.5
Residuals	B	D	4.85		
Total	C	E			

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Find values marked A, B, C, D, E, F and G in the ANOVA table.

1. A: 1

2. B: 7

3. C: 8

4. D: 33.95 (Keep all the decimal places in the answer)

5. E: 804.21 (Keep all the decimal places in the answer)

6. F: 770.26 (Keep all the decimal places in the answer)

7. G: 158.82 (Round off the answer up to the second decimal place)

(Type your answers within the given spaces)



Online Exams

Sri Lanka Institute of Information Technology

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lag question

A random group of 40 patients, who were suffered from high blood pressure, has average systolic blood pressure of 120 mmHg and standard deviation of 4.5. Construct a 90% confidence interval for the true mean of systolic blood pressure.

a) Lower Limit Equation : Choose... ▼

b) Upper Limit Equation : Choose... ▼

c) Critical value : Choose... ▼

d) Lower Limit value : Choose... ▼

e) Upper Limit value : Choose... ▼



Online Exams

Sri Lanka Institute of Information Technology

A Motor Company wants to investigate the average fuel consumption of a particular motor car it produces. The average fuel consumption is 16.5 km per one liter. New Technical staff of that Motor Company claims that the new average fuel consumption is 16.5 km per one liter. To test this, Motor company took the assistance of statistician and he selected a random sample of 10 cars. The fuel consumption of these cars is 16.5, 16.6, 16.7, 16.8, 16.9, 17.0, 17.1, 17.2, 17.3, 17.4 km/liter and 2.6. Test the claim of technical staff at 1% level of significance.

a) Hypothesis: $H_0: \mu = 16.5$ Vs $H_1: \mu \neq 16.5$

b) Test Statistic (Under H_0): $T = (\bar{X} - \mu) / (S / \sqrt{n-1})$

c) Distribution of test statistic: Normal Distribution ($N(16.5, 1)$)

d) Critical value: Choose...

e) Rejection criteria: Reject H_0 if $Z_{cal} > 2.58$ or $Z_{cal} < -2.58$ at 5% level of significance

f) Test value: Choose...

g) Conclusion (in Scientific term): Choose...

h) Conclusion (in general term): Choose...

Since $Z_{cal} = -1.70$ is in between -2.58 and 2.58 we do not reject H_0 at 1% level of significance
Since $Z_{cal} = -1.70$ is in between -1.64 and 1.64 we do not reject H_0 at 1% level of significance
Since $Z_{cal} = -2.1$ is in between -2.58 and 2.58 we do not reject H_0 at 5% level of significance
Since $Z_{cal} = 2.1$ is greater than 1.64, reject H_0 at 1% level of significance
Since $Z_{cal} = -1.70$ is in between -1.96 and 1.96 we do not reject H_0 at 1% level of significance



Question 9

Not yet answered

Marked out of 10.00

Flag question

A random group of 40 patients, who were suffered from high blood pressure, has a deviation of 4.5. Construct a 90% confidence interval for the true mean of systolic blood pressure.

a) Lower Limit Equation : Choose...

b) Upper Limit Equation : Choose...

c) Critical value : Choose...

d) Lower Limit value : Choose...

e) Upper Limit value : Choose...

- Choose...
- 122.47
- 120.12
- 126.22
- 121.85
- 121.43



Online Exams

Sri Lanka Institute of Information Technology

Question 3

Not yet answered

Marked out of 2.00

Flag question

In an experiment, it is given that $P(A) = 0.3$ and $P(B) = 0.5$ and $P(A \cup B) = 0.8$. Are A and B mutually exclusive?

Select one:

- ☒ Yes
- ☐ No
- ☐ Given information is not enough to decide.

Next page

Quiz navigation

DECLARATION

1

QUESTIONS

1 2 3 4 5 6 7 8

9 10

FEEDBACK

11

Ends attempt

Time left 13:08



Online Exams

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Question 8

Not yet answered

Marked out of 8.00

Flag question

After a survey, done by a Research group in a certain university in USA, claimed that 40% of the people in USA will vote for Donald Trump. From a sample of 50 people, without using any approximation, calculate the probability that

a) At least 20 people will vote for Donald Trump?

b) Fewer than 12 people vote for Donald Trump?

c) Using a suitable approximation, find the probability that less than 15 people will vote for Donald Trump?

Type your answers within the given spaces [Keep all decimal places of the final answer].



Question 9

Not yet answered

Marked out of 10.00

Flag question

A random group of 40 patients, who were suffered from high blood pressure, had a standard deviation of 4.5. Construct a 90% confidence interval for the true mean of systolic

a) Lower Limit Equation : Choose...

b) Upper Limit Equation : Choose...

c) Critical value : Choose...

d) Lower Limit value : 1.96

2.57

e) Upper Limit value : 2.01

-1.96

1.64



Question 9

Not yet answered

Marked out of
10.00

Flag question

A random group of 40 patients, who were suffered from high blood pressure, has average deviation of 4.5. Construct a 90% confidence interval for the true mean of systolic blood

- a) Lower Limit Equation : Choose...
Choose...
- b) Upper Limit Equation : $\bar{X} - t(0.10, 39) * (S/\sqrt{n})$
 $\bar{X} - Z(5\%) * (S/\sqrt{n})$
 $\bar{X} - Z(0.90) * (\sigma/\sqrt{n})$
 $\bar{X} - Z(10\%) * (S/\sqrt{n})$
 $\bar{X} - t(5\%, n) * (S/\sqrt{n})$
- c) Critical value : Choose...
- d) Lower Limit value : Choose...
- e) Upper Limit value : Choose...



Online Exams

Sri Lanka Institute of Information Technology

Question 10

Not yet answered

Marked out of 16.00

Flag question

A group of researchers wants to investigate the mean relief time given by a certain drug for Back Pain. They stated that the average relief time is 6.5 hours. But researchers stated that the relief time should be less than 6.5 hours. To claim, they selected 40 random patients who were suffering from Back Pain and given this drug. Average relief time was 5.8 hours with 2.25 standard deviation. Test the hypothesis at 5% level of significance.

a) Hypothesis : Choose...

b) Test Statistic (Under H_0) : Choose...

c) Distribution of test statistic : $Z = (X \text{ bar} - \mu) / (S / \sqrt{n})$

d) Critical value

e) Rejection criteria

f) Test value : Choose...

g) Conclusion (In Scientific term) : Choose...

h) Conclusion (In general term) : Choose...

Question 10

Not yet answered

Marked out of
16.00

Flag question

A group of researchers wants to investigate the mean relief time given by a certain drug. It is stated that the average relief time is 6.5 hours. But researchers stated that the relief time is not as claimed, they selected 40 random patients who were suffering from Back Pain and given this drug for 6.5 hours with 2.25 standard deviation. Test the hypothesis at 5% level of significance.

- a) Hypothesis : Choose... ▼
- b) Test Statistic (Under H_0) : Choose... ▼
- c) Distribution of test statistic : Choose... ▼
- d) Critical value : Choose... ▼
- e) Rejection criteria : Choose... ▼
- f) Test value : Choose... ▼
- g) Conclusion (In Scientific term) : Choose... ▼
- h) Conclusion (In general term) : Choose... ▼

Choose...

T distribution with $df=38$

T Distribution (N (0,1))

Normal Distribution (N (0,1))

Normal Distribution (N (0 , 5.06))

Normal Distribution (N (7.5 , 1))



Question 9

Not yet answered

Marked out of 10.00

Flag question

A random group of 40 patients, who were suffered from high blood pressure, has average systolic blood pressure of 123.12 and standard deviation of 4.5. Construct a 90% confidence interval for the true mean of systolic blood pressure.

a) Lower Limit Equation : Choose...

b) Upper Limit Equation : Choose...

c) Critical value : Choose...

d) Lower Limit value : Choose...

e) Upper Limit value : Choose...

Choose...

123.77

124.02

123.12

123.01

125.04



Online Exams

Sri Lanka Institute of Information Technology

A typist of a certain Book Publishing Company can type 7 pages per day. With

a) The typist type more than 8 pages per day

b) The typist type fewer than 3 pages per day

c) Using a suitable approximation, find the probability that the typist type less

Type your answers within the given spaces [Keep all decimal places of the final

Residuals B 2019.0 E

Total 24 D F

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i. What can be concluded using the scatter plot? There is a positive linear relationship in between

ii. State the estimated regression equation. Estimated Arm Strength = 12.561 + 1.789 (Dynamic Lift)

iii. State in how much dynamic lift will change if arm strength increased by one unit. Dynamic lift

iv. Does slope of the Regression line is significant ($\beta \neq 0$)? Use $\alpha = 5\%$. Yes. Regression line is significant

v. Use the regression equation to predict dynamic lift if arm strength is 43.4 units. 17.2381 units

g question

a) Find k value (Round off to three decimal values)

: 0

b) Find the Expected value of X . (Round off to three decimal values) :

0

c) Find the variance of X ($V(X)$) : (Round off to three decimal values) :

0

d) Find $F_X(0.5)$: (Round off to three decimal values)

0

(Type the correct answers within the given spaces)



Online Exams

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3
answered
out of
question

A retail merchant in USA has conducted a survey to determine the relationship in between weekly sales and weekly advertising expenditure of a company. He has collected data for 12 weeks and weekly advertising expenditure (\$) and weekly sales (\$). Following is the ANOVA table for the fitted regression model.

Analysis of Variance Table

Response: Weekly Sales (\$)

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Weekly advertising expenditure	A	C	28893.5	G	0.0001818**
Residuals	B	8698.2	E		
Total	11	D	F		

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Find values marked A, B, C, D, E, F and G in the ANOVA table.

1. A :
2. B :
3. C : (Keep all the decimal places in the answer)
4. D : (Keep all the decimal places in the answer)
5. E : (Keep all the decimal places in the answer)
6. F :
7. G : (Round off the answer up to the fourth decimal point)

(Type your answers within the given spaces except for "F")



Online Exams

Sri Lanka Institute of Information Technology

on 1
answered
d out of
question

A manufacturer claims that a jar of peanut butter contains mean of 500 grams with a standard deviation of 15.5 grams. Assuming that content of jar of peanut butter follows a normal distribution, find the following probabilities.

1. Probability that jar of peanut butter contains at most 530 grams : (Keep all the decimal places in the answer)
2. Probability that jar of peanut butter contains in between 470 grams and 520 grams : (Keep all the decimal places in the answer)
3. At which content (grams), 78% of jars of peanut butter have more than that content? grams (Keep the answer with two decimal points)

(Type your answers within the given spaces)

Next page

Answered

it of

Question

A retail merchant in USA has conducted a survey to determine the relation between weekly advertising expenditure (\$) and weekly sales (\$). The data is as follows:

$$\sum x = 405 \quad \sum y = 5510 \quad \sum xy = 192775 \quad \sum (x^2) = 15275 \quad \sum (y^2) = 3086010$$

Where X is the weekly advertising expenditure and Y is the weekly sales.

Calculate Pearson's correlation coefficient between the two variables (Give your answer up to 4 decimal places)

(Type your answer within the given space)

Answer:

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i. What can be concluded using the scatter plot?

Choose...

ii. State the estimated regression equation.

Choose...

iii. State in how much weekly sales will change if weekly advertising expenditure increased by one unit.

Choose...

iv. Does slope of the Regression line is significant ($\beta \neq 0$)? Use $\alpha = 5\%$.

Choose...

v. Use the regression equation to predict the weekly sales if weekly advertising expenditure is 34\$.

Choose...

Choose...

66.5\$

460.22\$

530.43\$

171.83\$

10749.09\$

ASUS VivoBook

2. Distribution of test statistic: Choose...

3. Significance level:

4. Critical value: Choose... ▼

5. Rejection criteria: Choose...

6. Test value (Round off to the nearest Integer):

7. Conclusion (In Scientific term): Choose...

8. Conclusion (In General term):

Choose...

11
83
76
109
123



Online Exams

Sri Lanka Institute of Information Technology

Question 5

Not answered

0 out of 1

Flag question

A manufacturer of PVC pipes claims that inside diameters of PVC pipes produced by his company are approximately normally distributed with a mean of 12 inches and standard deviation of 2.3 inches. Find the following probabilities.

1. Probability that diameter of a PVC pipe is at most 10 inches : (Keep all the decimal places)
2. Probability that diameter of a PVC pipe is in between 11 inches and 14 inches : (Keep all the decimal places)
3. At which diameter, 43% of PVC pipes have less than that diameter? inches (Keep the answer to two decimal places)

(Type your answers within the given spaces)

2. Distribution of test statistic:

Choose...

3. Significance level:

4. Critical value:

Choose... ▼

5. Rejection criteria:

Choose...

6. Test value (Round of

Choose...

7. Conclusion (In Scien

Reject H_0 if $X^2_{cal} < 18.0261$ at 1% level of significance

Reject H_0 if $X^2_{cal} > 9.48773$ at 5% level of significance

Reject H_0 if $X^2_{cal} > 9.48773$ at 0.5% level of significance

Reject H_0 if $t_{cal} > 4.604$ at 2.5% level of significance

Reject H_0 if $t_{cal} < 2.776$ at 10% level of significance

8. Conclusion (In General term):

Choose...

		450	285	2057
C's	568	83	47	698
D's and F's	85	15	3	103
Total	1975	548	335	2858

1. Hypothesis:

Choose...

2. Distribution of test statistic:

Choose...

3. Significance level:



4. Critical value:

Choose...

5. Rejection criteria:

Choose...

6. Test value (Round off to the nearest integer):



7. Conclusion (In Scientific term):

Choose...

8. Conclusion (In General term):

Choose...

Answered
t of
Question

The following table shows data for grades usually achieved in school and how often the respondent puts on sunscreen for more than 1 hour. Respondents are 12th-grade participants in the 2003 Youth Risk Behavior Survey sponsored by the U.S. Centers for Disease Control and Prevention, is a national survey of high school students. Test the association between sunscreen use and grade. Consider 5% level of significance.

Grade	Sunscreen Use			Total
	Never or rarely	Sometimes	Always or Most times	
A's and B's	1322	450	285	2057
C's	568	83	47	698
D's and F's	85	15	3	103
Total	1975	548	335	2858

1. Hypothesis:

Choose...

2. Distribution of test statistic:

Choose...

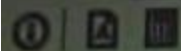
3. Significance level:

4. Critical value:

Choose...

5. Decision criteria:

Choose...



Online Exams

Sri Lanka Institute of Information Technology

A researcher reports that mice will live an average of 45 months when their diets are sharply restricted and proteins. Assuming that the lifetimes of such mice are normally distributed with a standard deviation of 7, a given mouse will live.

1. More than 30 months : (Keep all the decimal places in the answer)
2. Between 38 and 47 months : (Keep all the decimal places in the answer)
3. What is the life time (in months) of a mouse where 20% of all mice have less than this life-time? (to the nearest integer)

Type your answers within the given spaces)

Grade	Never or rarely	Sometimes	Always or Most times	Total
A's and B's	1322	450	285	2057
C's	568	83	47	698
D's and F's	85	15	3	103
Total	1975	548	335	2858

1. Hypothesis:

Choose...

Choose...

2. H0: There is no association in between sunscreen usage and grade obtained by respondents Vs. H1: There is an association in between sunscreen usage and grade obtained by respondents are dependent Vs. H1: Sunscreen usage and grade obtained by respondents are independent
3. H0: There is an association in between sunscreen usage and grade obtained by respondents Vs. H1: There is no association in between sunscreen usage and grade obtained by respondents are not related
4. H0: Sunscreen usage and grade obtained by respondents are related Vs. H1: Sunscreen usage and grade obtained by respondents are not related
5. H0: There is a relationship in between sunscreen usage and grade obtained by respondents Vs. H1: There is no relationship in between sunscreen usage and grade obtained by respondents

4. Critical value: Choose...

5. Rejection criteria: Choose...

6. Test value (Round off to the nearest integer):

7. Conclusion (In Scientific term): Choose...

8. Conclusion (In General term):

Choose...

Next page

Total

11

D

F

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i. What can be concluded using the scatter plot?

Choose...

ii. State the estimated regression equation.

Choose...

iii. State in how much weekly sales will change if weekly advertising expenditure increases by 1 unit.

Choose...

Choose...

Weekly sales will not change.

Weekly sales will decrease by 316.025\$.

Weekly sales will decrease by 4.241\$.

Weekly sales will increase by 4.241\$.

Weekly sales will increase by 316.025\$.

significant ($\beta \neq 0$)? Use $\alpha = 5\%$.

Choose...

What is the change in weekly sales if weekly advertising expenditure increases by 1 unit?



Online Exams

Sri Lanka Institute of Information Technology

Question 1

Not yet answered

Marked out of 16.00

Flag question

Past experience indicates that the time required for high school seniors to complete a standardized test (exam) is with an average of 35 minutes and standard deviation of 4.7 minutes. If a random sample of 20 high school seniors takes 38 minutes to complete this exam, test the hypothesis that average completion time (μ) has decreased, at the 0.05 level of significance.

a) Hypothesis : $H_0: \mu \geq 35$ Vs $H_1: \mu < 35$

b) Test Statistic (Under H_0) : $Z = (\bar{X} - \mu) / (\sigma / \sqrt{n})$

c) Distribution of test statistic : Normal Distribution ($N(0,1)$)

d) Critical value : 1.96

e) Rejection criteria : Reject H_0 if $Z_{cal} < -1.96$ at 5% level of significance

f) Test value : -11.04

g) Conclusion (in Scientific term) : Since $Z_{cal} = -11.04$ is less than -1.96, reject H_0 at 5% level of significance

h) Conclusion (in general term)

Therefore there is enough evidence to suggest that average completion time has decreased

	500	83	47	698
D's and P's	85	15	3	103
Total	1975	548	335	2858

1. Hypothesis:

Choose...

2. Distribution of test statistic:

Choose...

3. Significance level:

Choose...

4. Critical value:

Choose...

5. Rejection criteria:

Choose...

Chi squared distribution with 4 degrees of freedom
t distribution with 4 degrees of freedom
Chi squared distribution with 6 degrees of freedom
t distribution with 6 degrees of freedom
Chi squared distribution with 9 degrees of freedom

6. Test value (Round off to the nearest integer):

▼

7. Conclusion (In Scientific term):

Choose...

8. Conclusion (In General term):

Total

11

D

F

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i. What can be concluded using the scatter plot?

There is a positive linear relationship in between weekly sales and weekly advertising expenditure. ▼

ii. State the estimated regression equation.

Estimated Weekly sales = 316.025 + 4.241 (Weekly advert

iii. State in how much weekly sales will change if weekly advertising expenditure increased by one unit.

Weekly sales will increase by 4.241\$. ▼

iv. Does slope of the Regression line is significant ($\beta \neq 0$)? Use $\alpha = 5\%$.

Yes. Regression line is significant

v. Use the regression equation to predict the weekly sales if weekly advertising expenditure is 34\$. 4



Online Exams

Sri Lanka Institute of Information Technology

n3

answered

out of

g question

A study of the amount of rainfall and the quantity of air pollution removed produced a set of data. To examine the relationship between daily rainfall and particulate removed ($\mu\text{g}/\text{m}^3$), data were collected for 9 days. Following information is given:

$$\sum x = 45 \quad \sum y = 1094 \quad \sum xy = 5348.2 \quad \sum (x^2) = 244.26 \quad \sum (y^2) = 133786$$

Where X is the daily rainfall and Y is the particulate removed.

Calculate Pearson's correlation coefficient between the two variables (Give your answer in four decimal places).

(Type your answer within the given space)

Answer:

on 6

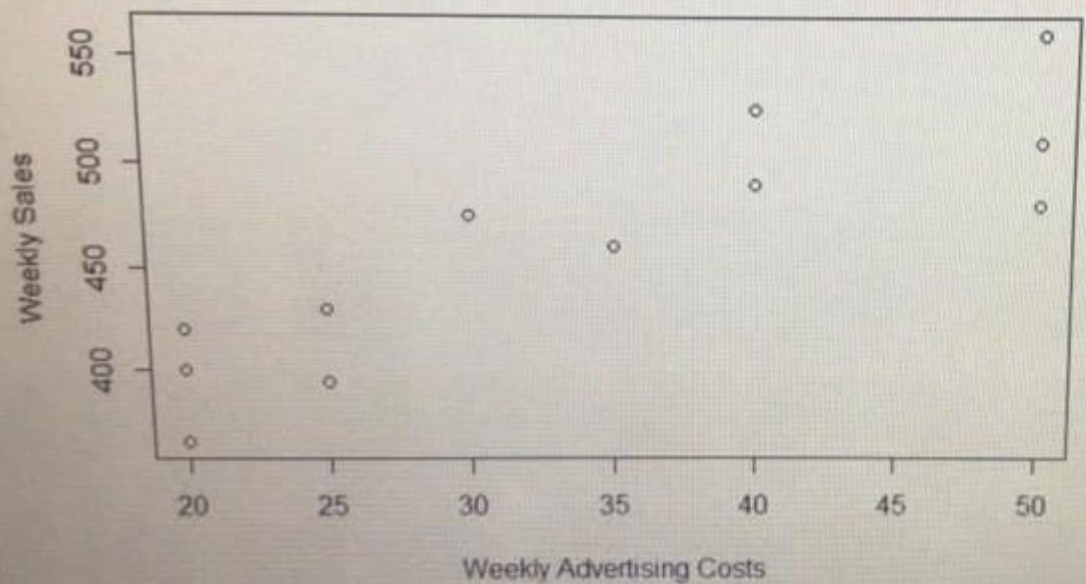
et answered

ed out of

lag question

A retail merchant in USA has conducted a survey to determine the relationship in between weekly ad company. He has collected data for 12 weeks and weekly advertising expenditure (\$) and weekly sales. Figure below displays the scatter plot for the data.

Scatter Plot for Weekly Sales Vs Weekly Advertising Costs



R outputs of the regression model are shown below.

Regression Model

Choose...

2. Distribution of test statistic:

Choose...

3. Significance level:

4. Critical value:

Choose... ▼

5. Rejection criterion:

Choose...

6. Test value (Round to nearest integer):

4.604

1.64

9.48773

18.0261

2.776

7. Conclusion (In S):

Choose...

8. Conclusion (In General term):

Choose...

1. Hypothesis:

Choose...

2. Distribution of test statistic:

Choose...

3. Significance level:



4. Critical value:

Choose...



5. Rejection criteria:

Choose...



6. Test statistic (Round off to the nearest integer):

Choose...

7.

Therefore there is enough evidence to suggest that sunscreen usage and grade obtained by responder

We can't give a conclusion since data is not enough.

Therefore there is enough evidence to suggest that there is no relationship in between sunscreen usage

Therefore there is enough evidence to suggest that there is an association in between sunscreen usage

8.

Therefore there is enough evidence to suggest that there is no association in between sunscreen usage

Choose...





Online Exams

Sri Lanka Institute of Information Technology

Question 8

Not answered

0 out of 10

Flag question

After a survey, done by a Research group in a certain university in USA, claimed that 40% of the people in the country will vote for Donald Trump. From a sample of 50 people, without using any approximation, calculate the probability that,

- a) At least 20 people will vote for Donald Trump?
- b) Fewer than 12 people vote for Donald Trump?
- c) Using a suitable approximation, find the probability that less than 15 people will vote for Donald Trump?

Type your answers within the given spaces [Keep all decimal places of the final answer].



Online Exams

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Question 4

Not answered

0 out of 10

Tag question

A Continuous random variable X has a probability density function given by.

$$f_X(x) = \begin{cases} k(1 - x^2) & ; -1 \leq x \leq 1 \\ 0 & ; \text{otherwise} \end{cases}$$

- a) Find k value (Round off to three decimal values) :
- b) Find the expected value ($E(X)$) . (Round off to 3 decimal values) :
- c) Find variance of X ($V(X)$). (Round off to three decimal values) :
- d) Find $F_X(0.5)$. (Round off to three decimal values) :

(Type the correct answers within the given spaces)

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i. What can be concluded using the scatter plot?

Choose...

ii. State the estimated regression equation.

Choose...

iii. State in how much weekly sales will change if weekly advertising expenditure increased by one unit.

Choose...

iv. Does slope of the Regression line is significant ($\beta \neq 0$)? Use $\alpha = 5\%$.

Choose...

v. Use the regression equation to predict the weekly sales if weekly advertising expenditure is 10 units.

Choose...

Data is not enough to test it.
No. Slope is equal to zero ($\beta = 0$).
Information is not enough to test it.
Yes. Regression line is significant.
No. Regression line is not significant.

Weekly advertising expenditure	A	C	28893.5	G	0.0001818***
Residuals	B	8698.2	E		
Total	11	D	F		

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i. What can be concluded using the scatter plot?

Choose...

ii. State the estimated regression equation.

Choose...

iii. State in how much weekly sales will change if weekly advertising expenditure increased by one unit.

Choose...

iv. Does slope of the Regression line is significant ($\beta \neq 0$)? Use $\alpha = 5\%$.

Choose...

v. Use the regression equation to predict the weekly sales if weekly advertising expenditure is 34\$.

Choose...

Residuals B D 4.85

Total C E

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i. What can be concluded using the scatter plot?

There is a negative linear relationship in between daily rainfall and particulate removed. ▼

ii. State the estimated regression equation.

Estimated Particulate Removed = $153.175 - 6.324$ (Daily Rainfall)

iii. State in how much particulate removed will change if daily rainfall increase by 1 mm.

Particulate removed will decrease by 153.175 units ▼

iv. Does slope of the Regression line is significant ($\beta \neq 0$)? Use $\alpha = 5\%$. Yes. Regression line is significant

v. Use the regression equation to predict particulate removed if daily rainfall is 6.4mm.

Choose...



Online Exams

Sri Lanka Institute of Information Technology

1

answered

out of

question

In an experiment, it is given that $P(A) = 0.3$ and $P(B) = 0.5$ and $P(A \cup B) = 0.8$. Are A and B independent?

Select one:

- ☐ Yes
- ☐ No
- ☐ Given information is not enough to decide.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Weekly advertising expenditure	A	C	28893.5	G	0.00018
Residuals	B	8698.2	E		
Total	11	D	F		

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i. What can be concluded using the scatter plot?

Choose...

Choose...

Data points in the plot are not enough to say anything about the plot.

There is no any relationship in between weekly sales and weekly advertising expenditure.

There is no pattern in the scatter plot.

There is a positive linear relationship in between weekly sales and weekly advertising expenditure.

There is a negative linear relationship in between weekly sales and weekly advertising expenditure.

iv. Does slope of the Regression line is significant ($\beta \neq 0$)? Use $\alpha = 5\%$. Choose...

v. Use the regression equation to predict the weekly sales if weekly advertising expenditure is



Question 1

Not yet answered

Marked out of
12.00

Flag question

A Continuous random variable X has a probability density function given by,

$$f_X(x) = \begin{cases} k(1 - x^2) & ; -1 \leq x \leq 1 \\ 0 & ; \text{otherwise} \end{cases}$$

a) Find k value (Round off to three decimal values) : 0.75

b) Find the expected value ($E(X)$) . (Round off to 3 decimal values) : .188

c) Find variance of X ($V(X)$). (Round off to three decimal values) : 0.165

d) Find $F_X(0.5)$. (Round off to three decimal values) : 0.563

(Type the correct answers within the given spaces)

g question

a) Find k value (Round off to three decimal values)

: 0

b) Find the Expected value of X . (Round off to three decimal values) :

0

c) Find the variance of X ($V(X)$) : (Round off to three decimal values) :

0

d) Find $F_X(0.5)$: (Round off to three decimal values)

0

(Type the correct answers within the given spaces)



Online Exams

Sri Lanka Institute of Information Technology

A researcher reports that mice will live an average of 45 months when their diets are sharp with vitamins and proteins. Assuming that the lifetimes of such mice are normally distributed with a standard deviation of 7.3 months, find the probability that a given mouse will live,

1. More than 30 months : (Keep all the decimal places in the answer)
2. Between 38 and 47 months : (Keep all the decimal places in the answer)
3. What is the life time (in months) of a mouse where 20% of all mice have less than this
(Give your answer to the nearest integer)

(Type your answers within the given spaces)

Regression Model

Coefficients

Intercept	Weekly Advertising Expenditure
316.025	4.241

Analysis of Variance

Response: Weekly Sales (\$)

	Df	Sum Sq	Mean Sq	F value
Weekly advertising expenditure	A	C	28893.5	G
Residuals	B	8698.2	E	
Total	11	D	F	

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Question 7

It has not yet been answered
Marked out of 1.00

Flag question

A retail merchant in USA has conducted a survey to determine the relationship in between weekly sales and weekly advertising expenditure (\$) of a company. He has collected data for 12 weeks and weekly advertising expenditure (\$) and weekly sales (\$). Following is the ANOVA table for the fitted regression model.

Analysis of Variance Table

Response: Weekly Sales (\$)

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Weekly advertising expenditure	A	C	28893.5	G	0.0001818***
Residuals	B	8698.2	E		
Total	11	D	F		

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Find values marked A, B, C, D, E, F and G in the ANOVA table.

1. A:

2. B:

3. C:

(Keep all the decimal places in the answer)

4. D:

(Keep all the decimal places in the answer)

5. E:

(Keep all the decimal places in the answer)

6. F:

7. G:

(Round off the answer up to the fourth decimal point)

(Type your answers within the given spaces except for "F")

Weekly advertising expenditure	A	C	2895.5	G	0.0001010
Residuals	B	8698.2	E		
Total	11	D	F		

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i. What can be concluded using the scatter plot?

Choose...

ii. State the estimated regression equation.

Choose...

iii. State in how much weekly sales will change

Choose...

Choose...
 Estimated Weekly sales = $316.025 - 4.241$ (Weekly adv
 Estimated Weekly sales = $4.241 + 316.025$ (Weekly adv
 Estimated Weekly advertising expenditure = $4.241 + 3$
 Estimated Weekly advertising expenditure = $316.025 +$
 Estimated Weekly sales = $316.025 + 4.241$ (Weekly adv

iv. Does slope of the Regression line is significant

v. Use the regression equation to predict the weekly sales if weekly advertising expenditure is 345.

Choose...

2. Distribution of test statistic: Choose...

3. Significance level: ▾

4. Critical value: Choose... ▾

5. Rejection criteria: Choose... ▾

6. Test value (Round off to the nearest integer): ▾

7. Conclusion (In Scientific term): Choose...

8. Conclusion (In General term):

Choose...

Choose...

Since $t_{cal} = 76 > 2.776$, reject H_0 at 0.5% level of significance
Since $X^2_{cal} = 83 > 9.48773$, reject H_0 at 5% level of significance
Since $X^2_{cal} = 109 > 9.48773$, do not reject H_0 at 2.5% level of significance
Since $t_{cal} = 11 > 4.604$, do not reject H_0 at 10% level of significance
Since $X^2_{cal} = 123 > 18.0261$, reject H_0 at 1% level of significance

hours with 2.25 standard deviation. Test the hypothesis at 5% level of significance.

a) Hypothesis

: $H_0: \mu \leq 6.5$ Vs $H_1: \mu > 6.5$ ▼

b) Test Statistic (Under H_0)

: $Z = (\bar{X} - \mu) / (\sigma / \sqrt{n})$ ▼

c) Distribution of test statistic

: Normal Distribution (N (0,1)) ▼

d) Critical value

: 1.64 ▼

e) Rejection criteria

: Choose... ▼

f) Test value

Choose...

There is enough evidence to suggest that average relief time is e

There is no enough evidence to suggest that average relief time i

There is enough evidence to suggest that average relief time is g

g) Conclusion (In Scientific term)

There is enough evidence to suggest that average relief time is le

There is enough evidence to suggest that average relief time is gr

h) Conclusion (In general term):

Choose...



IT2110 - Final Examination - Session 01 - Nov - 2020

IT2110 - Final Examination - Session 01 - Nov - 2020

This paper includes 10 main questions (5 MCQ questions and 5 Short answer questions) with sub questions. Duration is 2 hours. Backward navigation has disabled. Keep your Statistical table (Hard copy), Equation sheet (Hard copy) and Calculator with you before the exam starts.

There will be a text box at the end as the last question, asking any issues or errors in the quiz which is optional. If you think there is an error in a question, write that question number in that box (Better if you can give a short description).

This quiz has been configured so that students may only attempt it using the Respondus LockDown Browser.

Attempts allowed: 1

This quiz opened at Saturday, 28 November 2020, 11:30 AM

This quiz will close on Saturday, 28 November 2020, 2:00 PM.

Time limit: 2 hours

Summary of your previous attempts

State

Review

Finished

Submitted Saturday, 28 November 2020, 1:31 PM

No more attempts are allowed

[Back to the course](#)



Online Exams

Sri Lanka Institute of Information Technology

Question 2

Not yet answered

Marked out of 2.00

Flag question

A Continuous random variable X has a probability density function given by,

$$f_X(x) = \begin{cases} k(x^2 - 3) & ; -2 \leq x \leq 2 \\ 0 & ; \text{otherwise} \end{cases}$$

- a) Find k value (Round off your answer to three decimal values) :
- b) Find Expected Value (Round off your answer to three decimal values) :
- c) Find Variance (Round off your answer to three decimal values) :
- d) Find $F_X(1;2)$ (Round off your answer to three decimal values) :

(Type the correct answers within the given spaces)

Total C E

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

i. What can be concluded using the scatter plot?

There is a negative linear relationship in between year and average number of persons

ii. State the estimated regression equation.

Estimated Year = $44.39213 - 0.02095(A$

iii. State in how much average number of persons per household will change if one y

Average number of persons per household in US will decrease by 0.02095 units. ▼

iv. Does slope of the Regression line is significant ($\beta \neq 0$)? Use $\alpha = 5\%$.

Yes. Regress

v. Use the regression equation to predict average number of persons per household

Online Exams

Sri Lanka Institute of Information Technology

In an experiment, $P(A) = 0.4$ and $P(B) = 0.7$ and $P(A \cup B)' = 0.2$. Are A and B independent?

Select one:

- ☐ Yes
- ☐ No
- ☐ Given information is not enough to decide.

Coefficients

Intercept	Organic acid content
31.7087	0.3533

Analysis of Variance Table

Response: Acid content of material

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Organic acid content	A	C	2517.02	G	2.278e-05***
Residuals	B	262.98	E		
Total	9	D	F		

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- i. What can be concluded using the scatter plot? There is a positive linear relationship in between organic acid content and
- ii. State the estimated regression equation. Estimated Acid content of material = $31.7087 + 0.3533$ (Organic acid content)
- iii. State in how much acid content of material will change if organic acid content increased by one unit. Acid content of
- iv. Does slope of the Regression line is significant ($\beta \neq 0$)? Use $\alpha = 5\%$. Yes. Regression line is significant.
- v. Use the regression equation to predict the acid content of material if organic acid content is 143 units. 82.2306 units



Question 5

Not yet answered

Marked out of 3.00

Flag question

U.S. Census Bureau estimates of the average number of persons per household in the United States from 1850 to 2000 are shown in the following table. (These data are in the file perhouse on the computer.)

Year	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960	1970
Average number of persons per Household	5.55	5.28	5.09	5.04	4.93	4.76	4.54	4.34	4.11	3.67	3.37	3.35	3.14

To examine the relationship between the year and average number of persons per household in the United States, the following summary statistics were calculated:

$$\sum x = 30800 \quad \sum y = 65.15 \quad \sum xy = 124701.6 \quad \sum (x^2) = 59324000 \quad \sum (y^2) = 212.45$$

Where X is the year and Y is the average number of persons per household in the United States.

Calculate Pearson's correlation coefficient between the two variables (Give your answer to four decimal places.)

(Type your answer within the given space)

Answer:

er has found that average life time of a truck tire is 50500 miles with a standard deviation of 5000 miles. If the life time follows a normal distribution, find the following probabilities.

Probability that life time of a truck tire is at least 60000 miles : (Keep all the digits)

Probability that life time of a truck tire is 45000 to 58900 miles : (Keep all the digits)

What is the life time (in miles) where 38% of truck tires took less than this life time?

(er)

Give answers within the given spaces)

Online Exams

Sri Lanka Institute of Information Technology

A Manager of a certain super market is concerning about the number of customers arrive within the first hour. On average 9 customers are coming within the first hour. Without using any approximation,

- a) Fewer than 5 people are arriving to the Super Market.
- b) At least 6 people are arriving to the Super Market.
- c) Using suitable approximation, find the probability that more than 10 people are arriving to the Super Market.

Type your answers within the given spaces [Keep all decimal places of the final answer].



Online Exams

Sri Lanka Institute of Information Technology

A company, which produced candy products, required to test whether the average sugar content of a single toffee is different with the which is marked on the label. In the label it is marked as 2.5 g. From the previous studies it has been discovered that, sugar content has standard deviation of 0.15 and it is normally distributed. A group of researchers conducted an experiment to test this by taking a random sample of 28 toffees. Sample average sugar content was 2.4 g. Test the hypothesis at 5% level of significance.

a) Hypothesis : $H_0: \mu = 2.5$ Vs $H_1: \mu \neq 2.5$

b) Test Statistic : $Z = (\bar{X} - \mu) / (\sigma / \sqrt{n})$

c) Distribution of test statistic : Normal Distribution (N(0,1))

d) Critical value : Choose...

e) Rejection criteria : Choose...

f) Test value : Choose...

g) Conclusion (in Scientific term) : Choose...

h) Conclusion (in general term) : Choose...



Online Exams

Sri Lanka Institute of Information Technology

A researcher is interested in finding whether there is any relationship in between measured in degrees centigrade (°C). He used a sample size of 10 to study this relationship.

Analysis of Variance Table

Response: Proportion of impurities

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Temperature	A	0.65836	F	G	1.999e-05***
Residuals	B	D	0.00830		
Total	C	E			

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Find values marked A, B, C, D, E, F and G in the ANOVA table.

1. A:

2. B:

3. C:

4. D: (Keep all the decimal places in the answer)

5. E: (Keep all the decimal places in the answer)

6. F: (Keep all the decimal places in the answer)

7. G: (Round off the answer up to the fourth decimal point)

(Type your answers within the given spaces)



Online Exams

Sri Lanka Institute of Information Technology

A company, which produced candy products, required to test whether the average sugar content which is marked on the label. In the label it is marked as 2.5 g. From the previous studies it has been found that the standard deviation of 0.15 and it is normally distributed. A group of researchers conducted an experiment on a sample of 28 toffees. Sample average sugar content was 2.4 g. Test the hypothesis at 5% level of significance.

a) Hypothesis

: Choose...

b) Test Statistic

: Choose...

c) Distribution of test statistic

: Choose...

d) Critical value

: Choose...

e) Rejection criteria

: Choose...

f) Test value

: Choose...

g) Conclusion (in Scientific term):

Choose...

h) Conclusion (in general term):

Choose...

...past record say that students don't only go to campus. Following are the results.

Transportation mode	Drive	Bike	Other	Total
Frequency	100	150	50	300

Do the above results suggest that the past record for proportions have changed? Use 5% level of significance.

1. Hypothesis:

H_0 : Past record for mode of transportation hasn't changed. $\Pr(\text{drive}) = 0.3$, $\Pr(\text{Bike}) = 0.6$ and $\Pr(\text{other}) = 0.1$

2. Distribution of test statistic: Chi squared distribution with 2 degrees of freedom

3. Significance level: 5%

4. Critical value: 5.99146

5. Rejection criteria: Reject H_0 if $X^2 \text{ Cal} > 5.99146$ at 5% level of significance

6. Test value: 9.3285

7. Conclusion (in Scientific terms):

Since $t \text{ cal} = 12.5272 > 2.353$, do not reject H_0 at 10% level of significance

8. Conclusion (in General term):

Therefore there is enough evidence to suggest that there is at least one proportion which has changed after



Online Exams

Sri Lanka Institute of Information Technology

A company, which produced candy products, required to test whether the average sugar content of a single toffee is different with the value which is marked on the label. In the label it is marked as 2.5 g. From the previous studies it has been discovered that, sugar content has standard deviation of 0.15 and it is normally distributed. A group of researchers conducted an experiment to test this by taking a random sample of 38 toffees. Sample average sugar content was 2.4 g. Test the hypothesis at 5% level of significance.

- a) Hypothesis : $H_0: \mu \leq 2.5$ Vs $H_1: \mu > 2.5$
- b) Test Statistic : $Z = (\bar{X} - \mu) / (\sigma / \sqrt{n})$
- c) Distribution of test statistic : Normal Distribution ($N(0, 0.15)$)
- d) Critical value : ± 1.96 and -1.96
- e) Rejection criteria : Reject H_0 if $Z_{cal} = 1.96$ or $Z_{cal} < -1.96$ at 5% level of significance
- f) Test value : -3.53
- g) Conclusion (in Scientific term): Since $Z_{cal} = -3.53$ is less than -1.96 , we reject H_0 at 5% level of significance
- h) Conclusion (in general term) : There is enough evidence to suggest that the average sugar content is greater than to the value in the label.

DELL



Online Exams

Sri Lanka Institute of Information Technology

Question 10

Not yet answered

Marked out of 9.00

Flag question

A typist of a certain Book Publishing Company can type 7 pages per day. Without using any approximation, calculate the

a) The typist type more than 8 pages per day

b) The typist type fewer than 3 pages per day

c) Using a suitable approximation, find the probability that the typist type less than or equal to 4 pages per day.

Type your answers within the given spaces (Keep all decimal places of the final answer).

A random sample of 42 batteries, has on average life time of 88.5 hrs with 7.6 stan
the true mean lifetime (μ) of this brand of batteries.

a) Lower Limit Equation : Choose... ▼

b) Upper Limit Equation : Choose... ▼

c) Critical value : Choose... ▼

d) Lower Limit value : Choose... ▼

e) Upper Limit value : Choose... ▼



Question 3

Not yet answered

Marked out of 8.00

Flag question

A company, which produced candy products, required to test whether the average sugar content of a single toffee is different from the value which is marked on the label. In the label it is marked as 2.5 g. From the previous studies it has been discovered that, sugar content is normally distributed with a standard deviation of 0.15 and it is normally distributed. A group of researchers conducted an experiment to test this by taking a sample of 28 toffees. Sample average sugar content was 2.4 g. Test the hypothesis at 5% level of significance.

- a) Hypothesis : $H_0: \mu \leq 2.5$ Vs $H_1: \mu > 2.5$ •
- b) Test Statistic : $Z = (\bar{X} - \mu) / (\sigma / \sqrt{n})$ •
- c) Distribution of test statistic : Normal Distribution $N(0, 0.15)$ •
- d) Critical value : ± 1.96 and -1.96 •
- e) Rejection criteria : Reject H_0 if $Z_{cal} > 1.96$ or $Z_{cal} < -1.96$ at 5% level of significance •
- f) Test value : -3.53 •
- g) Conclusion (in Scientific term): Since $Z_{cal} = -3.53$ is less than -1.96 , we reject H_0 at 5% level of significance •
- h) Conclusion (in general term): There is enough evidence to suggest that the average sugar content is greater than the value marked on the label •

DELL



In an experiment, $P(A) = 0.4$ and $P(B) = 0.7$ and $P(A \cup B)^c = 0.2$. Are A and B independent?

Select one:

- ☐ Yes
- ☐ No
- ☐ Given information is not enough to decide.

A company, which produced candy products, required to test whether the average sugar content of a single toffee is different from 2.5 g. From the previous studies it has been discovered that, sugar content has standard deviation of 0.15 g. The company conducted an experiment to test this by taking a random sample of 28 toffees. Sample average sugar content was 2.4 g. Test the hypothesis.

a) Hypothesis : $H_0: \mu \leq 2.5$ Vs $H_1: \mu > 2.5$

b) Test Statistic : $Z = (\bar{X} - \mu) / (\sigma / \sqrt{n})$

c) Distribution of test statistic : Normal Distribution $(N(0,0.15))$

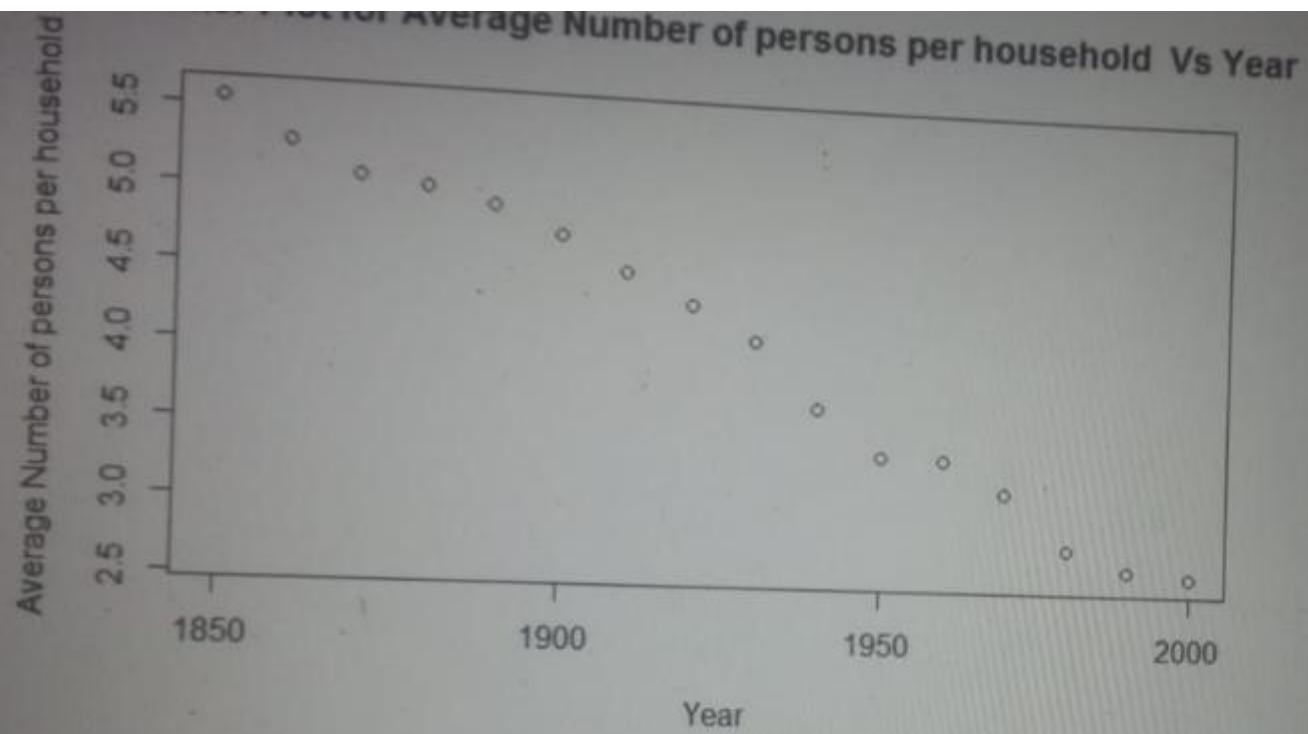
d) Critical value : +1.96 and -1.96

e) Rejection criteria : Reject H_0 if $Z_{cal} > 1.96$ or $Z_{cal} < -1.96$ at 5% level of significance

f) Test value : -3.53

g) Conclusion (In Scientific term): Since $Z_{cal} = -3.53$ is less than -1.96, we reject H_0 at 5% level of significance

h) Conclusion (In general term) : There is enough evidence to suggest that the average sugar content is greater than to the value of 2.5 g.



R outputs of the regression model are shown below.

Regression Model

Coefficients

Intercept	Year
44.39213	-0.02095

A retail merchant in USA has conducted a survey to determine the relationship in between weekly advertising expenditure (\$) and weekly sales (\$) have recorded for those 12 weeks. Follow

$$\sum x = 405 \quad \sum y = 5510 \quad \sum xy = 192775 \quad \sum (x^2) = 15275 \quad \sum (y^2) = 2567600$$

Where x is the weekly advertising expenditure and y is the weekly sales.

Calculate Pearson's correlation coefficient between the two variables (Give your answer in four decimal place)
(Type your answer within the given space)

Answer:



Online Exams

Sri Lanka Institute of Information Technology

Question 3

Not yet answered

Marked out of 3.00

Flag question

A researcher is interested in finding whether there is any relationship in between temperature and the proportion of through solid helium. Temperature is measured in degrees centigrade ($^{\circ}\text{C}$). He used a sample size of 10 to study this information is given.

$$\sum x = -2685.6 \quad \sum y = 7.007 \quad \sum xy = -1893.5644 \quad \sum (x^2) = 721454.96 \quad \sum (y^2) = 5.634547$$

Where X is the temperature and Y is the proportion of impurities.

Calculate Pearson's correlation coefficient between the two variables (Give your answer in four decimal places).

(Type your answer within the given space)

Answer: 0.9919



Online Exams

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A random sample of 45 metal bars has an average weight of 13.5 Kgs with 1.2 standard deviation. Find the 95% confidence interval for the true mean lifetime (μ) of this metal bars.

a) Lower Limit Equation: $\bar{X} - Z(2.5\%) * (S/\sqrt{n})$

b) Upper Limit Equation: $\bar{X} + Z(2.5\%) * (S/\sqrt{n})$

c) Critical value: Choose...

d) Lower Limit value: 1.96

e) Upper Limit value: -1.64

Choose...

1.96

1.64

-1.64

2.58

2.57

	≤ 4	5	6	≥ 7	
Abstainer	797	622	496	700	2615
Non-Binge	711	1139	984	2128	4962
Occasional Binge	247	443	471	1698	2859
Frequent Binge	167	448	360	2218	3193
Total	1922	2652	2311	6744	13629

1. Hypothesis:

H0: There is a relationship in between type of drinker and definition of binge drinking for men Vs. H1: There is no relationship in between type of drinker and definition of binge drinking for men

Choose...

H0: There is no relationship in between type of drinker and definition of binge drinking for men Vs. H1: There is a relationship in between type of drinker and definition of binge drinking for men

2. H0: Type of drinker and definition of binge drinking for men are dependent Vs. H1: Type of drinker and definition of binge drinking for men are independent

H0: There is an association in between type of drinker and definition of binge drinking for men Vs. H1: There is no association in between type of drinker and definition of binge drinking for men

3. H0: There is a relationship in between type of drinker and definition of binge drinking for men Vs. H1: There is no relationship in between type of drinker and definition of binge drinking for men

H0: Type of drinker and definition of binge drinking for men are related Vs. H1: Type of drinker and definition of binge drinking for men are not related

4. Critical value: Choose...

5. Rejection criteria: Choose...

6. Test value (Round off to the nearest Integer):

7. Conclusion (In Scientific term): Choose...

Online Exams

Sri Lanka Institute of Information Technology

Group of researchers have found that in a particular junction, on average, there will be 6 road accidents in a given month. W
approximation, calculate the probability that

- a) There will be at least 3 road accidents in a given month.
- b) Fewer than 2 road accidents in a given month.
- using a suitable approximation, find the probability that there will be at least 4 road accidents in a given month.
- owers within the given spaces (Keep all decimal places of the final answer).

Next Page



2

answered

out of

question

A random sample of 45 metal bars has an average weight of 13.5 Kgs with 1.2 standard deviation. Calculate the 95% confidence interval for the true mean lifetime (μ) of this metal bars.

a) Lower Limit Equation: $\bar{X} - Z(0.05) * (S/\sqrt{n})$

b) Upper Limit Equation: $\bar{X} + Z(0.05) * (S/\sqrt{n})$

c) Critical value: 1.64

d) Lower Limit value: 12.899

e) Upper Limit value: 13.765

Question 1

Not yet answered

Marked out of 7.00

Flag question

U.S. Census Bureau estimates of the average number of persons per household in the United States for each year from 1850 to 2000 are shown in the following table with the ANOVA output for the fitted regression model. (These data are from the U.S. Census Bureau's companion website.)

Year	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960	1970	1980	1990	2000
Average number of persons per Household	5.55	5.28	5.09	5.04	4.93	4.76	4.54	4.34	4.11	3.67	3.37	3.35	3.14	2.76	2.63	2.5

Analysis of Variance Table

Response: Average number of persons per household

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Year	A	14.9164	F	G	5.875e-14***
Residuals	B	D	0.0174		
Total	C	E			

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Find values marked A, B, C, D, E, F and G in the ANOVA table.

1. A: 2. B: 3. C: 4. D: (Keep all the decimal places in the answer)5. E: (Keep all the decimal places in the answer)6. F: (Keep all the decimal places in the answer)7. G: (Round off the answer up to the fourth decimal point)

(Type your answers within the given spaces)

Not yet answered
Marked out of 7.00
Flag question

2000 are shown in the following table with the ANOVA output for the fitted regression (companion website.)

Year	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960
Average number of persons per Household	5.55	5.28	5.09	5.04	4.93	4.76	4.54	4.34	4.11	3.67	3.37	3.35

Analysis of Variance Table

Response: Average number of persons per household

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Year	A	14.9164	F	G	5.875e-14***
Residuals	B	D	0.0174		
Total	C	E			

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Find values marked A, B, C, D, E, F and G in the ANOVA table.

1. A:
2. B:
3. C:
4. D: (Keep all the decimal places in the answer)
5. E: (Keep all the decimal places in the answer)
6. F: (Keep all the decimal places in the answer)
7. G: (Round off the answer up to the fourth decimal point)

(Type your answers within the given spaces)



Online Exams

Sri Lanka Institute of Information Technology

4

answered
out of

question.

A researcher is interested about a portion of a classic data set called the "pilot plant data" in Fitting Equations to Data by Daniel and Wood, published in 1971. The data set contains 10 data points. The response y is the acid content of material produced by titration, whereas the regressor x is the organic acid content produced by extraction and weighing. Following information is given.

$$\sum x = 999 \quad \sum y = 670 \quad \sum xy = 74058 \quad \sum (x^2) = 119900 \quad \sum (y^2) = 47670$$

Calculate Pearson's correlation coefficient between the two variables. Give your answer in four decimal places. (Type your answer within the given space).

Answer:

Next page

0.5

U.S. Census Bureau estimates of the average number of persons per household in the United States for census years between 1850 and 2000. ANOVA output for the fitted regression model. (These data are in the file perhouse on the companion website.)

Year	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960	1970	1980	1990	2000
Average number of persons per Household	5.55	5.28	5.09	5.04	4.93	4.76	4.54	4.34	4.11	3.67	3.37	3.35	3.14	2.76	2.63	2.59

Analysis of Variance Table

Response: Average number of persons per household

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Year	A	14.9164	F	G	5.875e-14***
Residuals	B	D	0.0174		
Total	C	E			

Signif. Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Find values marked A, B, C, D, E, F and G in the ANOVA table.

1. A:
2. B:
3. C:
4. D: (Keep all the decimal places in the answer)
5. E: (Keep all the decimal places in the answer)
6. F: (Keep all the decimal places in the answer)
7. G: (Round off the answer up to the fourth decimal point)

(Type your answers within the given spaces)

Previously, 70% come to campus in some other way (e.g., by walk, take the bus, get a ride). The campus sponsors a "spare the air" day to encourage people not to drive to campus. After the program, they want to know whether these proportions have changed. To test this hypothesis, a random sample of 300 students on a particular day was asked how they got to campus. Following are the results.

Transportation mode	Drive	Bike	Other	Total
Frequency	100	150	50	300

Do the above results suggest that the past record for proportions have changed? Use 5% level of significance.

1. Hypothesis:

H₀: Past record for mode of transportation hasn't changed. (Pr(Drive) = 0.3, Pr(Bike) = 0.6 and Pr(Other) = 0.1) Vs. H₁: At least one proportion

2. Distribution of test statistic: Chi squared distribution with 2 degrees of freedom

3. Significance level: 5%

4. Critical value: 5.99146

5. Rejection criteria: Reject H₀ if χ^2 Cal > 5.99146 at 5% level of significance

6. Test value: 9.3285

7. Conclusion (in Scientific terms): Since χ^2 Cal = 9.3285 > 5.99146, do not reject H₀ at 5% level of significance

8. Conclusion (in General terms):

Therefore there is enough evidence to suggest that there is at least one proportion which has changed after the program.

Next page



Online Exams

Sri Lanka Institute of Information Technology

Question 6

Not yet answered
Marked out of 10.00

Flag question

A random sample of 25 Candy Bars has an average sugar content of 9.8 grams. From the past studies, it is known that the sugar content follows a normal distribution and has an actual standard deviation as 1.3. Construct 90% confidence interval for the mean sugar content of metal bars.

Lower Limit Equation :

$\bar{x} - Z(5\%) * (\sigma/\sqrt{n})$

Upper Limit Equation :

$\bar{x} + Z(5\%) * (\sigma/\sqrt{n})$

Critical value

1.64

Lower Limit value

9.374

Upper Limit value

10.226

30-

TAKING YOU TO A
NEW REALITY

Intel OPTANE



U.S. Census Bureau estimates of the average number of persons per household between 1850 and 2000 are shown in the following table. (These data are in the file

Year	1850	1860	1870	1880	1890	1900	1910	1920	1930	1940	1950	1960
Average number of persons per Household	5.55	5.28	5.09	5.04	4.93	4.76	4.54	4.34	4.11	3.67	3.37	3.35

To examine the relationship between the year and average number of persons per household used.

$$\sum x = 36800 \quad \sum y = 65.15 \quad \sum xy = 124701.6 \quad \sum (x^2) = 59324000 \quad \sum (y^2) = 42.34$$

Where x is the year and y is the average number of persons per household in the United States.

Calculate Pearson's correlation coefficient between the two variables (Give your answer to 4 decimal places).

(Type your answer within the given space)

Answer: -0.9234

1

1. Hypothesis:

H₀: There is a relationship in between size of a family and level of education attained

2. Distribution of test statistic: Chi squared distribution with 4 degrees of freedom

3. Significance level: 1%

4. Critical value: 13.2767

5. Rejection criteria: Reject H₀ if $\chi^2_{\text{Cal}} > 13.2767$ at 1% level of significance

6. Test value (Round off to the nearest integer): 9

7. Conclusion (in Scientific term): Since $\chi^2_{\text{Cal}} = 9 < 13.2767$, do not reject H₀ at 1%

8. Conclusion (in General term):

Therefore there is enough evidence to suggest that size of a family is dependent on it